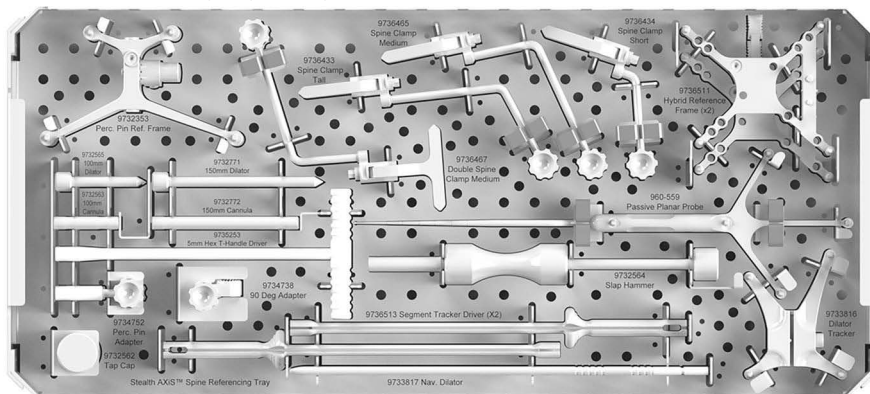


Stealth AXIS™ Spine Referencing Devices

All instruments described in this document may not be available in all geographies.

Included in the Spine Referencing Kit (9736691), the stainless steel Spine Referencing Kit (9736696), and the Advanced Spine Referencing Kit (9736692):



Standard 4-inch Bottom Tray Components:

- 100 mm Dilator (9732565)
- 100 mm Cannula (9732563)
- Percutaneous Pin Adapter (9734752)
- 150 mm Dilator (9732771)
- 150 mm Cannula (9732772)
- Slap Hammer (9732564)
- Double Spine Clamp Medium (9736467)
- 5 mm Hex T-Handle Driver (9735253)
- Tap Cap (9732562)
- 90 Degree Adapter (9734738)
- Passive Planar Probe (960-559) *
- Spine Clamp Short (9736434), Medium (9736465), Tall (9736433)
- Dilator Tracker (9733816) *
- Navigated Dilator (9733817) *
- Percutaneous Pin Reference Frame (9732353)
- Hybrid Reference Frame (9736511) * Not CE-marked at this time

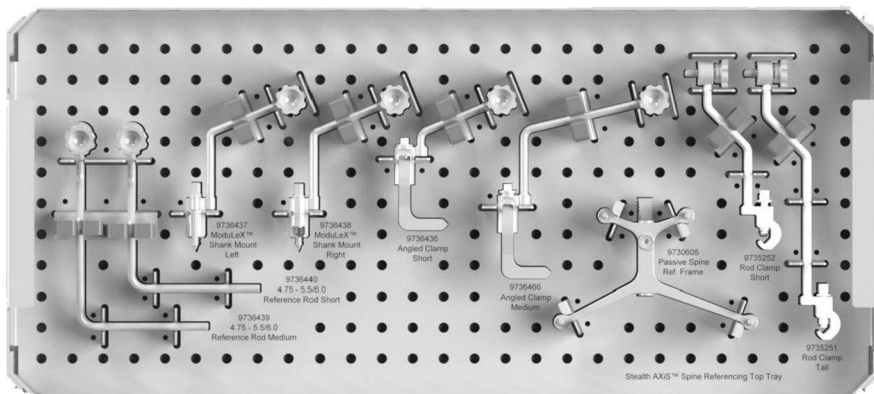
Optional 4-inch Tray Components:

One Hybrid Reference Frame (9736511) as an addition to the standard tray. Two frames may be stored on top of one another.

Two EM Segment Tracker Drivers (9736513) * as an addition to the standard tray. Two drivers may be stored in the tray. Not CE-marked at this time.

* Information regarding these devices is not included in this document. Please refer to the *Passive Planar Probe Package Insert*, *Hybrid Reference Frame Package Insert*, *EM Segment Tracker Driver Package Insert*, and *MAST Dilator Package Insert* for information on using these devices.

Only Included in the Advanced Spine Referencing Kit (9736692 Full Two Level Kit):



Standard 2-inch Top Tray Components:

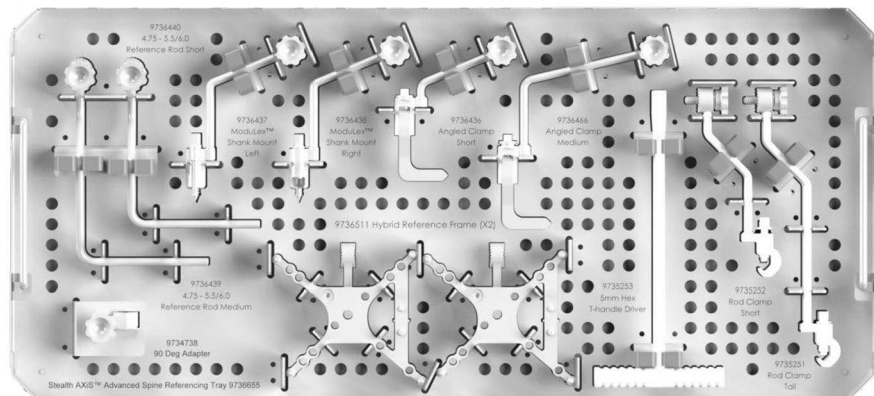
- Angled Clamp Short (9736436), Medium (9736466)
- Reference Rod Short 4.75-5.5/6.0 (9736440), Medium 4.75-5.5/6.0 (9736439)
- Rod Clamp Short (9735252), Tall (9735251)
- Passive Spine Reference Frame (9730605) *

Optional 2-inch Tray Components:

ModuLex™ Shank Mount Left (9736437), Right (9736438)

* Information regarding these devices is not included in this document. Please refer to the *Small Passive Frame Package Insert* for information on using these devices.

Only Included in the stainless steel Advanced Spine Referencing Kit (9736697):



Standard 2-inch Top Tray Components:

- Angled Clamp Short (9736436), Medium (9736466)
- Reference Rod Short 4.75-5.5/6.0 (9736440), Medium 4.75-5.5/6.0 (9736439)
- Rod Clamp Short (9735252), Tall (9735251)
- 5 mm Hex T-Handle Driver (9735253)
- 90 Degree Adapter (9734738)
- Hybrid Reference Frame (9736511) * Not CE-marked at this time

Optional 2-inch Tray Components:

ModuLex™ Shank Mount Left (9736437), Right (9736438)

One Hybrid Reference Frame (9736511) as an addition to the standard tray. Two frames may be stored next to each other.

* Information regarding these devices is not included in this document. Please refer to the *Hybrid Reference Frame Package Insert* for information on using these devices.

Intended Use / Indications for Use

When used with Medtronic computer-assisted surgery systems, defined as including the Stealth™ Surgical System, the following indications of use are applicable:

- The spine referencing devices are intended to provide rigid fixation between the patient and patient reference frame for the duration of the surgery. The devices are intended to be reusable.
- The percutaneous pin devices are intended to facilitate percutaneous placement and removal of a pin into the bony patient anatomy for rigid attachment of a patient reference. The devices are used during image-guided surgeries.
- The navigated instruments are specifically designed for use with Medtronic computer-assisted surgery systems, which are indicated for any medical condition in which the use of stereotactic surgery may be appropriate or vertebra can be identified relative to a CT or MR based model, fluoroscopy images, or digitized landmarks of the anatomy.
- The ModuLeX™ shank mounts are indicated to be used with the CD Horizon™ ModuLeX™ Spinal System during surgery.
- The Stealth™ spine clamps and ModuLeX™ shank mounts are indicated for skeletally mature patients.

Description

Spine referencing fixation devices provide rigid fixation to apply a patient reference frame. Various types of fixation devices are available:

Types of Spine Clamps (6 total):

- Straight Spine Clamps are available in Short (9736434), Medium (9736465), and Tall (9736433) versions.
- Angled Spine Clamps are available in Short (9736436) and Medium (9736466) versions.
- Double Spine Clamp is available as a Medium (9736467) version.

Other Devices:

- Reference Rods that can connect to two adjacent screw heads of diameters 4.75 to 6.0 mm and are available in Short (9736440) and Medium (9736439) versions.
- Rod Clamps are available in Short (9735252) and Tall (9735251) versions and are compatible with rod diameters ranging from 4.75 to 6.35 mm.
- Optional ModuLeX™ Shank Mounts are available in Left (9736437) and Right (9736438) versions. These Shank Mounts are compatible with any size ModuLeX™ shank.
- Percutaneous Pin Adapter (9734752) allows standard starburst connections to a Sterile Percutaneous Reference Pin.
- T-handle driver (9735253) is a tool used to attach/detach fixation devices including the spine clamps, rod clamps, and ModuLeX™ shank mounts.
- Cannulas, Dilators, the Tap Cap, and the Slap Hammer are only used when inserting or removing Sterile Percutaneous Reference Pins and are not fixed to the reference frame.

All referencing devices incorporate standard starburst style connection mechanisms to allow the use of current spine reference frames (including the Passive Spine Reference Frame [9730605], the Small Active Frame [9733886], and Hybrid Reference Frame [9736511]). The Hybrid Reference Frame may be used as a standard reference frame where standard registration techniques used in navigated surgeries are performed.

Contraindications

None.

Compatibility

All spine referencing devices are compatible with Medtronic computer-assisted surgery systems, including Stealth™ Surgical Systems and/or Mazor™ Surgical Systems, and Medtronic navigation instruments.

Material Composition

Material contained in product that can cause an allergic reaction: nickel

Patients can be exposed to the following material in these devices which are carcinogenic, mutagenic, and toxic for reproduction (CMR).

For additional materials of concern information such as REACH, CA Prop 65, or other product stewardship programs, go to www.medtronic.com/productstewardship.

System Level Manuals

These devices can be used with Medtronic computer-assisted surgical systems. See the Medtronic computer-assisted surgery system-level manuals and their associated pocket guides for descriptions of patient groups, intended users, clinical benefits, side effects, and residual risk disclosure statements. These devices are accessories to your Medtronic computer-assisted surgery system.

Warnings

- Warning:** The spine referencing devices have not been tested for use in pediatric patients. Use of these devices in these patients may result in poor fixation and device migration, which could result in navigation errors during surgery.
- Warning:** The spine referencing devices are designed for reuse and are supplied nonsterile. Clean and sterilize them before every use according to the instructions in this document.
- Warning:** The Spine Referencing Fixation Devices are not MR (magnetic resonance) qualified.
- Warning:** During navigation, frequently confirm navigational accuracy by touching the tip of the instrument on known anatomical landmarks and comparing the position of the instrument tip in the image with its physical location. Do not use the device if system inaccuracy is detected.
- Warning:** Visually inspect the device before each use. Do not use the device if it appears to be damaged, bent, or broken. Contact Medtronic Navigation.
- Warning:** When using the StealthAIR™ Spine registration method, place the device and reference frame within 30 cm (12 in) of the surgical area of interest for the registration process.
- Warning:** Do not move the patient in relation to the reference frame after image acquisition and registration. Moving the patient in relationship to the reference frame after image acquisition can cause inaccurate navigation during surgery. If a reference frame is suspected to have moved, a re-registration must be performed.
- Warning:** Do not use the Hybrid Reference Frame in conjunction with other patient reference frames.
- Warning:** Use caution and exercise care in order to prevent injury during use.
- Warning:** Abandon use if device damage or malfunction occurs during a procedure.
- Warning:** Avoid damage to these devices. Bending, prying, excessive force or misuse may result in breakage, failure of the instrument or localization errors, resulting in possible harm to the patient or the user.
- Warning:** Do not open sterile-barrier containers until surgical use. At time of use, inspect barrier for breach. If the sterile barrier was breached before surgical use, reprocess all devices contained in the container. If the sterile barrier for the Percutaneous Reference Pin is breached before surgical use, dispose of this device.
- Warning:** Do not attempt to alter the product. Only use the product in accordance with its labeling.
- Warning:** Do not re-use or re-process a device where transmissible spongiform encephalopathy agents or Creutzfeldt-Jakob disease (prion) contamination is suspected or confirmed. Incinerate the potentially contaminated device according to national guidelines.
- Warning:** Prior to using devices in the Spine Referencing and Advanced Spine Referencing Kits, inspect the devices for cleanliness, damage, or deterioration. Do not attempt to use the devices if any of the components appear to be bent or otherwise damaged.
- Warning:** The Percutaneous Reference Pin is provided sterile and comes in two lengths: 100 mm Disposable Pin (9733235) and a 150 mm Disposable Pin (9733236). This device is for single use only. Do not reprocess, resterilize, or reuse this device. Attempts to reprocess or resterilize the device may be ineffective and may compromise its structural integrity. Any reuse creates a risk of contamination which could result in patient injury, illness, or death.
- Warning:** The sterile spheres are for single use only. Do not re-process, re-sterilize, or re-use the sterile spheres.
- Warning:** Abandon use of the sterile devices if their sterility is compromised.
- Warning:** Other kit components are not provided sterile. Before use, clean and sterilize the kit components according to the instructions in this document.
- Warning:** Touching metal instruments to electrocautery may expose user to harmful electric current.
- Warning:** To avoid potential exposure to blood-borne pathogens and chemicals, use appropriate Personal Protective Equipment when handling, reprocessing, or disposing of Medtronic devices. Universal precautions for handling this instrument after use should be observed by all hospital personnel according to OSHA Standard 29 CFR 1910.1030, Occupational Exposure to Blood borne Pathogens.
- Warning:** Remove all Cannulas and pins from the patient when removing the device.
- Warning:** Do not change the angle of the patient reference after registration.
- Warning:** Avoid bumping or moving the reference frame relative to the patient's position. If the frame is suspected to have moved during a procedure, re-register the patient to confirm navigational accuracy.
- Warning:** Disengage all components from the reference frame before cleaning and sterilizing.

Precautions

- Caution:** Federal law (U.S.A.) restricts these devices to sale by or on the order of a physician.
- Caution:** This device has not been tested for use in patients with compromised integrity of the spinous process or pedicle stability due to bone deficiencies such as, but not limited to, osteoporosis, osteopenia, tumors, or skeletal deformities. Use of these devices in these patients may result in poor fixation, device migration, or navigation errors during surgery, which could lead to loss of spinal stabilization, neurological injury, or the need for revision surgery.

Caution: Refer to the Medtronic computer-assisted system-level manuals and associated pocket guides for additional instructions, warnings, and cautions.

Caution: Do not throw or drop the device. Throwing or dropping a reference device may cause damage to it.

Caution: During navigation with the devices, visually confirm navigational accuracy frequently by localizing on known points and comparing the image of the instrument to its location in physical space.

Caution: These devices are designed and tested for use only with Medtronic instruments. Only use these devices in conjunction with supported Medtronic products.

Straight, Double, and Angled Spine Clamp Use

1. Use the T-handle driver to loosen the drive screw and open the jaws of the clamp.
2. Position the clamp over the spinous process of surgical interest. The Straight Spine Clamps should be centered over one spinous process, the Double Spine Clamp should be centered over two adjacent spinous processes, and the Angled Spine Clamps should be centered over spinous processes that require longitudinal fixation to affix properly. The straight clamps, double clamp, and angled clamps all have a clamping range from 0 mm (fully closed) to 11 mm (fully open).
3. Holding the clamp stationary, use the T-handle driver to tighten the adjustment screw until the clamp is securely affixed to the bony anatomy.
4. Partially attach a spine reference frame to the assembly with the Starburst Connector Screw. Orient frame so that the 4 spheres are visible to the camera's line of sight (does not have to be perpendicular). If necessary, use the 90 Degree Adapter (9734738) between the frame and the clamp to gain a better angle to the system camera.
5. Fully tighten the Starburst Connector Screw, and verify that the reference frame is secure. Make sure that the reference frame is visible to the Medtronic computer-assisted surgery system localization camera when positioned for surgery.
6. To remove the clamp, use the T-handle driver to loosen the adjustment screw while supporting the clamp by the post until the clamp jaws are free of the bony anatomy.

Rod Clamp Use

The Rod Clamp is intended for rod diameter sizes of 4.75 through 6.35 mm in diameter.

1. Use the T-handle driver to loosen the adjustment screw on the hook end of the clamp so that it is able to fit around the pedicle rod.
2. Position the hook end of the clamp onto an appropriate position of the pedicle rod.
3. Holding the clamp stem stationary, use the T-handle driver to tighten the adjustment screw until the clamp is securely affixed to the pedicle rod. Make sure that the clamp is rigidly secured to the rod.
4. Partially attach a spine reference frame to the assembly with the Starburst Connector Screw. Orient the frame so that the 4 spheres are visible to the camera's line of sight (does not have to be perpendicular). If necessary, use the 90 Degree Adapter (9734738) between the frame and the clamp to gain a better angle to the system camera.
5. Fully tighten the Starburst Connector Screw, and verify that the reference frame is secure. Make sure that the reference frame is visible to the Medtronic computer-assisted surgery system localization camera when positioned for surgery.
6. To remove the clamp, use the T-handle driver to loosen the adjustment screw until the clamp is free of the rod.

Reference Rod Use

The Reference Rod is for use with Medtronic screw heads that accept rod sizes of 4.75 to 6.0 mm in diameter only. Do not implant the device. Make sure that the Reference Rod is always used with two pedicle screw heads. Do not attach the reference to only one screw head.

1. Position the Reference Rod onto an appropriate position between two pedicle screw heads.
2. Use a set screw and the appropriate screwdriver from the Medtronic screw set to anchor the Reference Rod into each screw head. Make sure that the rod is rigidly secured to the screw head.
3. Break off the set screw head to ensure the rigidity of the device.
4. Attach a spine reference frame to the Reference Rod using the Starburst Connector Screw, but don't fully tighten the screw. Orient the frame so that the 4 spheres are visible to the camera's line of sight (does not have to be perpendicular). If necessary, use the 90 Degree Adapter (9734738) between the frame and the clamp to gain a better angle to the system camera.
5. Fully tighten the Starburst Connector Screw, and verify that the reference frame is secure. Make sure that the reference frame is visible to the Medtronic computer-assisted surgery system localization camera when positioned for surgery.
6. To remove the Reference Rod, loosen and remove the set screws from the screw heads until the Reference Rod is free. Set screws should be removed to avoid an accidental impact. The Reference Rod must be removed and is not intended to be implanted.

ModuLex™ Shank Mount Use

The ModuLex™ Shank Mounts are intended to be used with a ModuLex™ shank. Do not implant the device.

1. Position the ModuLex™ shank mount over the selected shank and slide over the spherical portion of the shank.

2. Holding the clamp stem stationary by the post, use the T-handle driver to tighten the adjustment screw until the shank mount is securely affixed to the shank. Ensure that the shank mount is in line with the shank. Off-axis alignment will lead to insecure attachment.
3. To prevent the risk of the shank rotating, use the T-handle driver to drive the stabilization screw into bone, being mindful of sensitive anatomy.
Note: The adjustment screw is used to secure the shank mount to the shank. This is separate from the stabilization screw, which stabilizes the shank and shank mount.
4. Attach a spine reference frame to the shank mount using the Starburst Connector Screw, but don't fully tighten the screw. Orient the frame so that the 4 spheres are visible to the camera's line of sight (does not have to be perpendicular). If necessary, use the 90 Degree Adapter (9734738) between the frame and the clamp to gain a better angle to the system camera.
5. Fully tighten the Starburst Connector Screw, and verify that the reference frame is secure. Make sure that the reference frame is visible to the Medtronic computer-assisted surgery system localization camera when positioned for surgery.
6. To remove the mount, use the T-handle driver to retract the stabilization screw and loosen the adjustment screw until the mount can slide off the shank.

Percutaneous Reference Pin and Adapter Use

Setup

1. Select the appropriate size Percutaneous Reference Pin, 100 mm (9733235) or 150 mm (9733236), that reaches the bone through the surrounding tissues ensuring rigid fixation to the bone and a proper connection to the reference frame.
2. Make a small incision in the area over the pelvis.
3. Place the corresponding size Dilator in the Cannula. Insert the assembly into the incision and push it through the tissue until it touches bone.
4. Remove the Dilator while holding the Cannula in place.
5. Insert the pin through the Cannula.
6. Place the Tap Cap on the hind end of the pin. Ensure that the arrow on the top of the Tap Cap faces the system camera and that the pin guide slides all the way up the slot in the Tap Cap.
7. Use a mallet to nail the pin into bone until the Tap Cap bottoms out on the Cannula.
Note: The measuring tick marks present on the side of the Sterile Percutaneous Reference Pin are accurate within 0.25 mm. These marks are meant to be used only in conjunction with the Cannula and the Tap Cap. When the pin is docked into the entry point on the bone, the top of the Cannula aligns with the 0 tick mark. As the pin is hammered into the bone using the Tap Cap, the tick marks and Cannula together serve as a depth gauge up to 25 mm, when the Tap Cap contacts the Cannula.
8. Remove the Tap Cap from the pin.
Note: Make sure the Sterile Percutaneous Reference Pin is rigidly anchored in bone. If rigid fixation is not possible, other Medtronic methods are available.

Option A, steps 9-11 only valid when using the Percutaneous Pin Adapter (9734752):

9. Carefully place the Percutaneous Pin Adapter onto the anchored Percutaneous Reference Pin by sliding the end of the adapter onto the hind end of the pin and turning the adapter clockwise until it stops.
10. Partially attach a spine reference frame (can use either the Hybrid Reference Frame [9736511] or the Passive Spine Reference Frame [9730605]) to the assembly with the Starburst Connector Screw. Orient the frame so that the 4 spheres are visible to the camera's line of sight (does not have to be perpendicular). If necessary, use the 90 Degree Adapter (9734738) between the frame and the clamp to gain a better angle to the system camera.
11. Fully tighten the Starburst Connector Screw, and verify that the reference frame is secure. Make sure that the reference frame is visible to the Medtronic computer-assisted surgery system localization camera when positioned for surgery.

Option B steps 12-13 only valid when using the Percutaneous Pin Reference Frame (9732353):

12. The Percutaneous Pin Reference Frame is placed on the Sterile Percutaneous Reference Pin until the post on the side of the pin meets the top of the working channel of the reference frame. The reference frame is then rotated until it is locked into position.
13. Secondary adjustment of the reference frame is achieved by engaging a spring-loaded button and tilting the frame in the cranial or caudal direction.
Valid when using either Option A or B:
14. Verify that the system displays the correct reference frame on screen. If necessary, select the Small Passive Frame in the application software.

Removing the Sterile Percutaneous Reference Pin

1. Detach the patient reference from the Sterile Percutaneous Reference Pin by pressing down on the reference and turning it counter-clockwise.
2. Remove the Percutaneous Pin Adapter by pushing it down and turning it counter clockwise to free the adapter from the pin. Pull the adapter straight off the pin.
3. Slip the end of the Slap Hammer onto the hind end of the pin.

4. Hold the pin, Cannula, and the tip of the Slap Hammer with one hand. Use your other hand to slide the hammer along the shaft away from the bone in swift oscillations until the pin is dislodged from the bone.
5. Properly dispose of the pin.
6. Remove the Cannula from the incision.

Reference Frame Use

The Percutaneous Pin Reference Frame (9732353) is only compatible with a Sterile Percutaneous Reference Pin. The Hybrid Reference Frame (9736511), the Small Active Frame (9733886), and the Passive Spine Reference Frame (9730605) are compatible for use with the following devices:

- Spine Clamp Short (9736434), Medium (9736465), and Tall (9736433)
- Double Spine Clamp Medium (9736467)
- Angled Spine Clamp Short (9736436), Medium (9736466)
- 90 Degree Adapter (9734738)
- Rod Clamp Short (9735252), Tall (9735251)
- Reference Rod Short (9736440), Medium (9736439)
- ModuLeX™ Shank Mount Left (9736437), Right (9736438)
- Percutaneous Pin Adapter (9734752)
- Existing referencing fixation devices using the starburst type attachment mechanism and any type of Medtronic Navigation spine patient registration or intraoperative image acquisition.

Warning: Do not bump or move the reference fixation device after patient registration. If the fixation device is suspected to have moved, users must check navigation accuracy. If inaccurate, perform a re-registration.

Warning: If remounting the reference frame or verifying an instrument, check navigation accuracy. If inaccurate, perform a re-registration.

Warning: If registration has already occurred, check navigation accuracy after verifying an instrument with a reference frame that is attached to the patient.

Warning: Passive reflective spheres are single use disposable products. Do not attempt to reprocess, resterilize, or reuse any single use product.

Caution: When using StealthAiR™ Spine Registration, the device can only be used with a CT or CT-like device with a minimum field of view of 30 cm (12 in).

Caution: The device is to be used with a minimum CT bore size of 40 cm (16 in).

Note: The devices listed above are also compatible with the Small Active Frame (9733886). For indications for use, instructions, warnings, and cautions related to the use of the Active Frame, see the *Small Active Frame Package Insert*.

Standard Registration Technique

1. Before attaching the reference frame, place passive reflective spheres on the patient reference frame's 4 sphere mounting stems. Make sure that the spheres are fully seated on the mounting stem. Keep the spheres clean during use.
2. Make sure the reference frame is securely attached to the Starburst Connector Screw on the fixation device within the area of surgical interest for the standard registration.
3. Ensure the reference frame can be seen by the Medtronic computer-assisted surgery system's localization camera when the patient is positioned for surgery.
4. Perform the standard registration process for the procedure. Please see the Medtronic computer-assisted surgery system-level manuals and their associated pocket guides.
5. Perform accuracy checks after registration by localizing on known anatomical landmarks within the area of interest.

Standard Instrument Verification Technique

1. Ensure the reference frame can be seen by the Medtronic computer-assisted surgery system's localization camera.
2. Make sure that the instrument's tool card is selected in the Medtronic computer-assisted surgery system spine software.
3. Place the instrument tip straight into the reference frame divot and press the footswitch or wait until the system verifies/calibrates the instrument.
4. The instrument can now be used in the procedure.

Spine Referencing Device Reprocessing - Cleaning, Inspection, and Sterilization

Warning: The recommended sterilization parameters are only valid with CE-marked equipment that is properly maintained and calibrated.

Caution: Use the rigid container for sterilization only. Instruments must be cleaned separate from the container.

Spine Referencing Device Cleaning

Cleaning is the initial step and sterilization occurs later in reprocessing.

Cleaning is the removal of organic soil. Effective cleaning:

- Minimizes organic soil transfer from one patient to another.
- Prevents accumulation of residual soil throughout the product's useful life.
- Facilitates successful sterilization. Adequate reprocessing is contingent upon the thoroughness of cleaning.

* Information regarding these devices is not included in this document. Please refer to the *Passive Planar Probe Package Insert*, *Small Passive Frame Package Insert*, *Hybrid Reference Frame Package Insert*, *EM Segment Tracker Driver Package Insert*, and *MAST Dilator Package Insert* for information on cleaning and sterilizing these devices.

Warning: Use Personal Protective Equipment (PPE) as necessary when handling, processing, or disposing of the device. Universal precautions should be observed by all personnel according to OSHA Standard 29 CFR 1910.1030 Occupational Blood borne Pathogens.

Limitations on Reprocessing	Warning: Prior to use, examine the device components for damage, deterioration, deformation, and abuse. Do not attempt to use any instrument that appears to be bent or otherwise damaged. Abandon the use of any device damaged during the procedure. See the Maintenance Inspection Testing instructions in this document to determine if the device is suitable for use. If defects are found, do not use the product. Contact Medtronic Navigation. Warning: Do not reprocess devices suspected of having prion contamination. If prion contamination is suspected or confirmed, incinerate the device according to national regulations. Caution: At the end of its useful life, dispose of the device in accordance with national regulations. The useful life of a device depends on a variety of factors including method and duration of each use, as well as handling between uses. End of useful life is normally determined by wear and damage due to use. Careful inspection and functional testing prior to use is the best method for determining if the device is at the end of its useful life. See the "Maintenance Inspection Testing" section in this document to determine if the device is at the end of its useful life.
Point of Use	Reprocessing begins at the point of use. Do not allow blood, debris, or bodily fluids to dry on the devices. Remove excess soil using cold running tap water, or wipe the device with a clean, lint-free cloth.
Containment and Transportation	Warning: To prevent cross-contamination, always handle, transport, clean, sterilize, and package devices contacting the central nervous system separately from other devices. Caution: Devices should be cleaned within 30 minutes of use to limit fixation of contaminants. Caution: To prolong the life of the device, reprocess immediately after use. If the device cannot be reprocessed immediately, keep the device moist during transport.

Preparation for Cleaning	Caution: Remove or detach all devices. Remove any single-use spheres and dispose of them in accordance with national regulations. No special tools are required: grasp the sphere at its base and pull. Caution: Do not use metal brushes, scouring pads, or abrasive cleaners as this may damage the device. • Thoroughly rinse the device under running cold tap water at a temperature of 10-22°C (50-72°F) to remove any visible soil. While rinsing, actuate any movable surfaces through their full range of motion. Use a soft-bristled nylon brush and/or a clean lint-free cloth to aid in soil removal. Special attention should be given to lumens, crevices, and other features that may present a challenge to cleaning. Continue to rinse, brush, and actuate the movable surfaces under cold running tap water for a minimum of 1 minute. • Flush any lumens with the running cold tap water. Use a soft-bristled nylon brush of appropriate diameter and length to clean the lumen such that the bristles contact the entire inner diameter of the lumen and moves through the entire length of the lumen while maintaining grasp to pull back through. Push the lumen brush through one end of the lumen until it exits the other end. Pull the lumen brush back through the lumen until it exits the device. Rinse the brush and repeat until the brush is not removing visible soil from inside the lumen. • Carefully inspect devices, including any lumens and cavities, to ensure all visible soil is removed. If any soil remains on the device, continue rinsing and brushing the device until it is visibly clean.
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Automated Cleaning (Enzymatic)	• Transfer the devices to the washer-disinfector, placing instruments at an angle to allow for draining and to ensure device cavities are facing down, and then select the instrument cycle. • Ensure the following cycle parameters are properly programmed. Note: Enzymatic cleaner and detergent manufacturers provide concentrated cleaners and detergents. The enzymatic cleaners and detergents must be diluted per the manufacturer's instructions.			
	Phase	Recirculation/ Soak (minutes)	Water Temperature	Cleaner/Detergent Type
	Prewash	2 minutes (02:00)	Cold tap water 10-22°C (50-72°F)	N/A
	Enzyme Wash	2 minutes (02:00)	Hot tap water 43-55°C (109-131°F)	Neutral pH enzymatic cleaner
	Wash	2 minutes (02:00)	66°C (151°F) (set-point)	Neutral pH detergent
	Rinse	15 seconds (00:15)	Hot tap water 43-60°C (109-140°F)	N/A
	Thermal Rinse	1 minute (01:00)	90°C (194°F) (set-point)	N/A
	Purified Water Rinse	10 seconds (00:10)	66°C (151°F) (set-point)	N/A
	Warning: After cleaning, visually examine all parts of the device for visible soil. If visible soil is observed, repeat cleaning. Note: Cycle validated using Steris Valsure™ Enzymatic Cleaner and Steris Valsure™ Neutral Detergent or Steris Prolystica™ 2x Concentrate Enzymatic Cleaner and Neutral Detergent.			

Automated Cleaning (Alkaline)	- Transfer the devices to the washer-disinfector, placing instruments at an angle to allow for draining and to ensure device cavities are facing down, and then select the instrument cycle. - Ensure the following cycle parameters are properly programmed. Caution: The Small Passive Frame (9730605), Percutaneous Pin Frame (9732353), Dilator Tracker (9733816), and the Passive Planar Probe (960-559) can only be cleaned using enzymatic (neutral) cleaning solutions. Certain solutions (i.e. alkaline based) may significantly damage the devices. These solutions must not be used. Note: Alkaline detergent manufacturers provide concentrated alkaline detergents. The alkaline detergents must be diluted per the detergent manufacturer's instructions. * Alkaline cleaning does not apply to devices described in the <i>Passive Planar Probe Package Insert</i> and <i>MAST Dilator Package Insert</i>.			
	Phase	Recirculation/ Soak (minutes)	Water Temperature	Detergent Type
	Prewash	2 minutes (02:00)	Cold tap water 10-22°C (50-72°F)	N/A
	Alkaline Wash	5 minutes (05:00)	Hot tap water 43°C (109°F)(setpoint)	Alkaline detergent
	Rinse	1 minute (01:00)	Hot tap water 43-60°C (109-140°F)	N/A
	Thermal Rinse	1 minute (01:00)	90°C (194°F) (setpoint)	N/A
	Purified Water Rinse	1 minute 30 seconds (01:30)	66°C (151°F) (setpoint)	N/A
	Warning: After cleaning, visually examine all parts of the device for visible soil. If visible soil is observed, repeat cleaning. Note: Cycle validated using Dr. Weigert neodisher® MediClean forte.			
Drying	If necessary, dry the device with a clean, lint-free towel.			
Maintenance Inspection Testing	Warning: Do not reprocess for surgical use a device that has obvious damage or corrosion. Visually inspect the device for obvious damage or corrosion to ensure it is not pitted, fractured, bent, or otherwise damaged. Make sure of the following: - All device components are accounted for. - Laser etchings, engravings, and other markings are legible. - There is no damage to the device. The device should be free from chips, cracks, sharp-edged gouges, bending, wearing, and other damage. - The starburst connection is secure. - The Starburst Connector Screw is free from damage (nicks, gouges, bends, etc.) that would interfere with the mating function. - Discoloration, corrosion, stains or rust are not present. If so, attempt to clean using the cleaning instructions provided in this document. If reprocessing does not remove the corrosion, stains, or rust, the device is at the end of its useful life. - All parts are present and free of damage and deterioration. - There is no damage to the working ends or tips of the percutaneous pin. The working end should be free from cracks, sharp-edged gouges, and other damage. - There is no damage to the threads. - Mating ends are free of damage (nicks, gouges, bends, etc.) that would interfere with the mating function. A device that shows or exhibits properties listed above is at the end of its useful life. Dispose of the device according to national regulations.			

Packaging	Warning: Do not wrap rigid sterilization containers. Devices may not be sterilized to an adequate Sterility Assurance Level (SAL) if the container is wrapped. Caution: The device sterilization tray is to be used only for sterilization, not for cleaning. Package the devices for sterilization using one of the following configurations: • In a Medtronic branded sterilization inner instrument tray within a Medtronic branded reusable rigid sterilization container (9736632, 9736654, 9736633, or 9736655) with disposable filters, such as SCF01 paper filters. • Individually double-wrapped in sterilization wrap, such as Kingaurd KC600 Sterilization Wrap. Use this method only if a reusable rigid sterilization container and sterilization instrument tray are not available. • The inner trays include graphical outlines of the devices as well as corresponding part numbers for precise placement. Note: Some instrument slots within the trays may be intentionally empty. Contact Medtronic Navigation for questions regarding device placement within the trays. Note: Always make sure that clamps are in a closed position for placement in the sterilization container.
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Spine Referencing Device Sterilization

Caution: Do not sterilize and supply for surgical use any device that is not visibly clean and free of particulates. If particulates are present, repeat reprocessing, starting with the Preparation for Cleaning step.

Caution: If all devices were not removed or detached prior to sterilization, remove or detach all devices and repeat the cleaning process. Remove any single-use spheres and dispose of them in accordance with national regulations. No special tools are required: grasp the sphere at its base and pull.

Caution: Because of the variability in cleaning efficiencies and sterilizer operating parameters, all given parameters (temperature, time, etc.) should be validated by persons who have training and expertise in sterilization processes. Deviation from the recommended sterilization processes is at the risk of the user facility.

Caution: Device cannot be sterilized to an adequate Sterility Assurance Level (SAL) without prior cleaning and decontamination.

Caution: Do not stack containers or trays during sterilization as this may limit steam penetration and prevent effective sterilization.

Note: The single-use Sterile Percutaneous Reference Pins are EO (ethylene-oxide) sterilized.

Sterilization is intended to kill microorganisms, reduce the likelihood of transmission, and reduce the possibility of infection.

1. Load the trays and container into the sterilizer following the sterilizer manufacturer's load procedures and load configurations.
2. Program the sterilizer using the parameters below for medical facilities in accordance with the table headings.

Sterilization parameters

Sterilization	Use a prevacuum (dynamic air removal) steam (moist heat) sterilization cycle with these parameters to sterilize the device.			
	Table 1: Sterilization cycle parameters for the medical facilities in any geography			
	Cycle	Temperature	Exposure Time	Minimum Dry Time*
	Prevacuum	132°C (270°F)	4 minutes (04:00)	30 minutes (30:00)
	Prevacuum	135°C (275°F)	3 minutes (03:00)	30 minutes (30:00)
	Table 2: Sterilization cycle parameters for medical facilities outside the United States of America and its territories			
	Cycle	Temperature	Exposure Time	Minimum Dry Time*
	Prevacuum	134°C (273°F)	3 minutes (03:00)	30 minutes (30:00)
	Prevacuum	134°C (273°F)	18 minutes (18:00)	30 minutes (30:00)
	*Minimum dry times were validated using sterilizers having vacuum drying capabilities. Drying cycles using ambient atmospheric pressures may require more time. The time required to dry the device may vary with environmental conditions. Warning: Steam for sterilization should be generated from water that has been treated to remove total dissolved solids, filtered to remove contaminants and water droplets, and supplied via piping without deadlegs or other stagnant zones where contamination might collect. Steam saturation should be greater than 97%.			
Storage	- The devices should remain in their rigid sterilization container, sterilization instrument tray, or sterilization wrap. Store in cool, dry, and clean conditions at ambient room temperature such that the package is not compromised. - The shelf-life of a device depends on the sterile barrier system, storage manner, and environmental conditions. Shelf-life is dependent upon the legally marketed reusable rigid sterilization container or sterilization wrap. The health care facility should establish a shelf-life for sterilized instrumentation based on the type of packaging used and the recommendations of the sterile barrier system manufacturer.			

Disposal

Dispose of contaminated devices according to your facility's biohazard disposal procedure and local and national regulations.

Reporting

If a serious incident occurs in relation to the use of these devices, report it to Medtronic Navigation. If this incident occurs in the European Union, also report it to the competent authority in the Member State where the incident occurred.

Assistance

If you have questions about using spine referencing devices, contact your local Medtronic Navigation representative or call Medtronic Navigation, Inc. at 800 595 9709 or 720 890 3200.

Symbol definitions

The following symbols may appear on spine referencing device packaging:

R_X Only **Caution:** Federal law (U.S.A.) restricts this device to sale by or on the order of a physician.



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